TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	70.1 ± 5.28
Correlation Stability	84.4 ± 6.41
Distribution Consistency	69.7 ± 5.55
Entropy Preservation	77.3 ± 11.04
Mean Z Score Drift	0.6 ± 0.29
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	99.5 ± 0.50
Feature Accuracy	52.4 ± 20.54
Novelty Score	0.9 ± 0.04

TabuLLM: AI Benchmark Report

Dataset Audit: heart_failure_clinical_records.csv

1. Dataset Portrait

Attribute	Value
Rows	299
Columns	13
Numeric features	13
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	60.834	63.454	+2.620	4.3%	OK
anaemia	0.431	0.368	-0.063	14.7%	Warn
creatinine_phosphokinase	581.839	1923.816	+1341.976	230.6%	Drift
diabetes	0.418	0.247	-0.171	41.0%	Drift
ejection_fraction	38.084	35.992	-2.092	5.5%	OK
high_blood_pressure	0.351	0.268	-0.083	23.7%	Warn
platelets	263358.029	327165.297	+63807.268	24.2%	Warn
serum_creatinine	1.394	4.629	+3.235	232.1%	Drift
serum_sodium	136.625	128.749	-7.876	5.8%	OK
sex	0.649	0.820	+0.171	26.4%	Drift
smoking	0.321	0.167	-0.154	47.9%	Drift
time	130.261	132.598	+2.337	1.8%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	70.1 ± 5.28
Correlation Stability (*2)	84.4 ± 6.41
Distribution Consistency (*3)	69.7 ± 5.55
Entropy Preservation (*4)	77.3 ± 11.04
Mean Z Score Drift (*5)	0.6 ± 0.29
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	99.5 ± 0.50
Feature Accuracy (*10)	52.4 ± 20.54
Novelty Score (*11)	0.9 ± 0.04

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6133333333333335	0.635	0.6299999999999999	0.01666666666666653	
F1	0.007142857142857143	0.0899835528867787	0.07867370011298047	0.07153084297012333	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8433333333333334	0.78	0.8116666666666668	-0.03166666666666676	
F1	0.7697701062711032	0.623630283732484	0.7083835318440254	-0.06138657442707935	*

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7499999999999999	0.7683333333333333	0.7699999999999999	0.01999999999999973	
F1	0.5667171896628354	0.6278301475080738	0.5639090054399081	-0.00280818422292718	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7900000000000001	0.7133333333333333	0.7983333333333333	0.008333333333333326	
F1	0.6814301967147093	0.5708757133326178	0.7077944827896754	0.02636428607496027	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6333333333333334	0.64	0.635	0.001666666666666607	
F1	N/A	0.08108178053830227	0.03375757575757575	0.03375757575757575	

Model: AB

TabuLLM: AI Benchmark Report

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.825	0.755	0.82	-0.005000000000000044	
F1	0.7417404771442427	0.563586436289238	0.7166597773219652	-0.025080699822277543	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8283333333333334	0.7683333333333333	0.8266666666666665	-0.0016666666666666718	
F1	0.7384740266630495	0.579491378796635	0.7275168826835945	-0.010957143979454886	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8300000000000001	0.7699999999999999	0.8266666666666668	-0.0033333333333333327	
F1	0.7532401666570763	0.6236658367891803	0.7367337950092709	-0.016506371647805253	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.805	0.7833333333333333	0.805	1.1102230246251566e-17	
F1	0.6916473027290337	0.6438500819640116	0.6833462131175047	-0.008301089611529078	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5116666666666667	0.5183333333333333	0.5199999999999999	0.008333333333333331	
F1	0.2936061348598413	0.19798982820144945	0.2393396721693986	-0.05426646269044273	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
2	55.0	0.0	7861.0	0.0	38.0	0.0	263358.03	1.1	136.0	1.0
3	65.0	0.0	146.0	0.0	20.0	0.0	162000.0	1.3	129.0	1.0
4	50.0	1.0	111.0	0.0	20.0	0.0	210000.0	1.9	137.0	1.0
5	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
6	90.0	1.0	47.0	0.0	40.0	1.0	204000.0	2.1	132.0	1.0
7	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0
8	60.0	1.0	315.0	1.0	60.0	0.0	454000.0	1.1	131.0	1.0
9	65.0	0.0	157.0	0.0	65.0	0.0	263358.03	1.5	138.0	0.0
10	80.0	1.0	123.0	0.0	35.0	1.0	388000.0	9.4	133.0	1.0
11	75.0	1.0	81.0	0.0	38.0	1.0	368000.0	4.0	131.0	1.0
12	62.0	0.0	231.0	0.0	25.0	1.0	253000.0	0.9	140.0	1.0
13	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
14	50.0	1.0	168.0	0.0	38.0	1.0	276000.0	1.1	137.0	1.0
15	49.0	1.0	80.0	0.0	30.0	1.0	427000.0	1.0	138.0	0.0
16	82.0	1.0	379.0	0.0	50.0	0.0	47000.0	1.3	136.0	1.0
17	87.0	1.0	149.0	0.0	38.0	0.0	262000.0	0.9	140.0	1.0
18	45.0	0.0	582.0	0.0	14.0	0.0	166000.0	0.8	127.0	1.0
19	70.0	1.0	125.0	0.0	25.0	1.0	237000.0	1.0	140.0	0.0
20	48.0	1.0	582.0	1.0	55.0	0.0	87000.0	1.9	121.0	0.0

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	53.0288510522385 2	1.0	582.0	1.0	41.0	0.0	233776.0	0.81545295730565 13	138.0	0.0
2	95.0	0.0	23.0	0.0	65.0	0.0	850000.0	9.4	148.0	1.0
3	95.0	0.0	23.0	0.0	65.0	0.0	850000.0	0.50846105631757 95	113.0	1.0
4	74.6890767171957	0.0	3269.0	0.0	52.0	0.0	216665.0	0.89406150373336	136.0	0.0
5	95.0	1.0	23.0	0.0	65.0	0.0	850000.0	9.4	113.0	1.0
6	40.0582787739533 05	0.0	23.0	0.0	14.0	0.0	25100.0	9.4	113.0	1.0
7	45.0	1.0	66.0	1.0	25.0	0.0	233000.0	0.8	135.0	1.0
8	70.0	0.0	161.0	0.0	25.0	0.0	244000.0	1.2	142.0	0.0
9	58.0	1.0	400.0	0.0	40.0	0.0	164000.0	1.0	139.0	0.0
10	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	40.0	0.0	23.0	1.0	14.0	0.0	25100.0	5.0	113.0	1.0
2	95.0	0.0	23.0	1.0	14.0	0.0	25100.0	5.0	113.0	1.0
3	95.0	1.0	23.0	0.0	80.0	0.0	25100.0	5.0	148.0	1.0
4	40.000218188889	0.0	23.0	1.0	14.0	0.0	25100.0	5.0	113.0	1.0
5	95.0	0.0	23.0	0.0	14.0	0.0	25100.0	5.0	113.0	1.0
6	40.0	0.0	23.0	1.0	14.0	0.0	25100.0	5.0	113.0	1.0
7	60.0	0.0	2656.0	1.0	30.0	0.0	305000.0	2.3	137.0	1.0
8	45.0	1.0	130.0	0.0	35.0	0.0	174000.0	0.8	139.0	1.0
9	45.0	0.0	582.0	1.0	55.0	0.0	543000.0	1.0	132.0	0.0
10	65.0	1.0	59.0	1.0	60.0	0.0	172000.0	0.9	137.0	0.0

TabuLLM: AI Benchmark Report

B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	40.0	0.0	23.0	0.0	80.0	0.0	730080.0	0.5000138397771261	147.0	1.0
2	40.799720624882994	1.0	85.0	0.0	35.0	0.0	307359.0	1.1476619383916058	133.0	1.0
3	57.87443292099285	0.0	377.0	0.0	38.0	0.0	312115.0	1.3051169856968656	137.0	1.0
4	75.84163306798591	0.0	929.0	0.0	41.0	0.0	225899.0	1.0	135.0	1.0
5	40.00827415593359	1.0	291.0	0.0	35.0	0.0	180045.0	0.9665147417912912	147.0	1.0
6	95.0	0.0	23.0	0.0	14.0	0.0	25100.0	9.4	121.0	1.0
7	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0
8	79.0	1.0	55.0	0.0	50.0	1.0	172000.0	1.8	133.0	1.0
9	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
10	70.0	0.0	92.0	0.0	60.0	1.0	317000.0	0.8	140.0	0.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	52.32274635044345	0.0	587.0	1.0	38.0	0.0	251953.0	0.8142554807704414	137.0	0.0
2	40.195447865567075	0.0	133.0	0.0	36.0	0.0	286249.0	1.1472552121649002	118.0	1.0
3	40.0	1.0	30.0	1.0	15.0	0.0	25100.0	0.6	148.0	1.0
4	74.70156145221992	0.0	976.0	0.0	45.0	0.0	212053.0	0.9269165365002646	136.0	0.0
5	40.10716461158295	1.0	242.0	1.0	35.0	0.0	184411.0	0.9696747845343096	144.0	1.0
6	94.0	0.0	7861.0	1.0	80.0	0.0	25100.0	6.1	113.0	1.0
7	48.0	1.0	131.0	1.0	30.0	1.0	244000.0	1.6	130.0	0.0
8	95.0	1.0	371.0	0.0	30.0	0.0	461000.0	2.0	132.0	1.0
9	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
10	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0

TabuLLM: AI Benchmark Report

B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	40.0	0.0	23.0	0.0	14.0	0.0	51000.0	0.5	148.0	1.0
2	95.0	1.0	23.0	1.0	70.0	1.0	51000.0	0.5210139675923176	113.0	0.0
3	95.0	1.0	7861.0	1.0	70.0	1.0	51000.0	0.5	148.0	0.0
4	73.46010245032755	0.0	3838.0	0.0	47.0	0.0	222772.0	0.8958369934629865	137.0	0.0
5	95.0	0.0	23.0	1.0	70.0	1.0	51000.0	9.4	113.0	1.0
6	65.54457915317491	0.0	95.0	0.0	35.0	1.0	228795.0	1.1479193976764488	134.0	1.0
7	78.0	0.0	224.0	0.0	50.0	0.0	481000.0	1.4	138.0	1.0
8	70.0	0.0	582.0	0.0	20.0	1.0	263358.03	1.83	134.0	1.0
9	61.0	0.0	248.0	0.0	30.0	1.0	267000.0	0.7	136.0	1.0
10	40.0	0.0	478.0	1.0	30.0	0.0	303000.0	0.9	136.0	1.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	40.0	0.0	23.0	0.0	80.0	0.0	742000.0	9.4	116.0	1.0
2	40.34804214441994	1.0	78.0	0.0	37.0	0.0	377986.0	1.1704193172762443	125.0	1.0
3	95.0	0.0	23.0	0.0	80.0	1.0	742000.0	9.4	148.0	0.0
4	76.32201281729182	1.0	1436.0	0.0	42.0	0.0	234268.0	1.0	138.0	1.0
5	41.11331496381741	1.0	463.0	1.0	36.0	0.0	167156.0	0.9952015481397065	145.0	1.0
6	40.10582422013240 5	1.0	23.0	0.0	80.0	1.0	25100.0	9.4	116.0	1.0
7	81.0	0.0	4540.0	0.0	35.0	0.0	231000.0	1.18	137.0	1.0
8	70.0	0.0	2695.0	1.0	40.0	0.0	241000.0	1.0	137.0	1.0
9	70.0	0.0	835.0	0.0	35.0	1.0	305000.0	0.8	133.0	0.0
10	80.0	0.0	776.0	1.0	38.0	1.0	192000.0	1.3	135.0	0.0

TabuLLM: AI Benchmark Report

B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	51.0283550143756 1	0.0	582.0	1.0	39.0	0.0	250933.0	0.86758024039332 13	137.0	0.0
2	40.0000320048651 76	1.0	72.0	0.0	35.0	0.0	350380.0	1.11591667994780 97	132.0	1.0
3	60.3658342782063 2	0.0	246.0	0.0	37.0	0.0	310707.0	1.24283191721065 46	138.0	1.0
4	74.2926240372762 4	0.0	826.0	0.0	43.0	0.0	214841.0	1.02448236951723 31	135.0	1.0
5	41.0314145454803 74	1.0	240.0	0.0	32.0	0.0	113064.0	1.03566457750091 41	146.0	1.0
6	60.1046982868695 2	1.0	105.0	0.0	15.0	0.0	33269.0	1.99624176323348 01	132.0	1.0
7	75.0	0.0	582.0	1.0	30.0	1.0	263358.03	1.83	134.0	0.0
8	53.0	0.0	207.0	1.0	40.0	0.0	223000.0	1.2	130.0	0.0
9	60.0	1.0	2281.0	1.0	40.0	0.0	283000.0	1.0	141.0	0.0
10	59.0	1.0	280.0	1.0	25.0	1.0	302000.0	1.0	141.0	0.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	50.2743114598813 5	1.0	598.0	1.0	40.0	0.0	237787.0	0.82458886487187 17	139.0	0.0
2	42.0991709881885 26	1.0	136.0	0.0	35.0	0.0	336875.0	1.0	132.0	1.0
3	95.0	1.0	30.0	0.0	80.0	1.0	25100.0	9.4	113.0	1.0
4	74.8859464775752 9	0.0	1161.0	0.0	44.0	0.0	234324.0	0.93370411158293 4	136.0	0.0
5	95.0	1.0	30.0	0.0	80.0	1.0	850000.0	9.4	113.0	1.0
6	95.0	1.0	30.0	0.0	80.0	1.0	25100.0	9.4	113.0	1.0
7	65.0	0.0	224.0	1.0	50.0	0.0	149000.0	1.3	137.0	1.0
8	80.0	0.0	898.0	0.0	25.0	0.0	149000.0	1.1	144.0	1.0
9	66.0	1.0	72.0	0.0	40.0	1.0	242000.0	1.2	134.0	1.0
10	75.0	1.0	582.0	0.0	30.0	0.0	225000.0	1.83	134.0	1.0

TabuLLM: AI Benchmark Report

B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	40.0	0.0	7702.0	0.0	14.0	0.0	47000.0	0.6	113.0	1.0
2	40.0048167756149 1	0.0	54.0	0.0	31.0	0.0	294970.0	1.62885017915791 73	117.0	1.0
3	95.0	1.0	23.0	0.0	65.0	0.0	47000.0	6.8	113.0	1.0
4	81.2869871990477 6	1.0	2785.0	0.0	41.0	0.0	205766.0	0.93407230598373 49	136.0	1.0
5	47.2778867545847 1	1.0	407.0	1.0	35.0	0.0	164435.0	1.03778955004463 01	140.0	1.0
6	95.0	1.0	23.0	0.0	65.0	0.0	47000.0	6.8	147.0	1.0
7	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0
8	60.0	0.0	68.0	0.0	20.0	0.0	119000.0	2.9	127.0	1.0
9	70.0	0.0	97.0	0.0	60.0	1.0	220000.0	0.9	138.0	1.0
10	53.0	1.0	1808.0	0.0	60.0	1.0	249000.0	0.7	138.0	1.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	95.0	0.0	23.0	0.0	14.0	0.0	25100.0	0.5	148.0	1.0
2	95.0	1.0	5882.0	0.0	80.0	1.0	850000.0	9.4	116.0	1.0
3	40.0	0.0	5882.0	0.0	80.0	0.0	850000.0	9.4	116.0	1.0
4	40.0	1.0	5882.0	0.0	80.0	1.0	850000.0	9.4	116.0	1.0
5	40.0	0.0	5882.0	1.0	80.0	0.0	850000.0	9.4	116.0	1.0
6	40.0	1.0	5882.0	0.0	80.0	0.0	850000.0	9.4	116.0	1.0
7	60.0	0.0	897.0	1.0	45.0	0.0	297000.0	1.0	133.0	1.0
8	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
9	45.0	1.0	1876.0	1.0	35.0	0.0	226000.0	0.9	138.0	1.0
10	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0